

Wellmere

AP Biology · heredity

A breeding station on a headland, ringed by isolation distance. Sea on three sides and one causeway in.

45	29	2.3	3/6	7	31 · 12 · 2
STOPS	CHOICE (64%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

The worst mix in the catalogue — 29 of 45 stops are CHOICE — on a syllabus whose centrepiece nothing computes. Phase 0 will clear two of the four gaps (the recombination fraction is computed on a DEGENERACY board; N_e is computed on a BALLPARK whose relationship the keyword list cannot see), leaving **two real gaps, both ratios, both on one stop**. Seven select-only concepts, five of them mechanisms.

NOW The mix as it ships

CHOICE 29	BALLPARK 3	SEQUENCE 2	DEGENERACY 1	CONTROL 1	HOLDOUT 1	TRIGGER 1	BALANCE 1	TRACE 1	SCIENCETANK 1	DELEGATE 1	VALUE 1	DIAGNOSIS 1	ALLOCATE 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
3 : 1	the monohybrid ratio, and what a count of a whole generation is evidence for	4 · CHOICE	real gap
15 : 1	two genes acting either-way, and the ratio that separates it from one	4 · CHOICE 5 · DEGENERACY	verify first
recombination frequency $\approx$ map distance in cM	linkage: how often a marker and a gene come apart	5 · DEGENERACY 42 · CHOICE	verify first

2 of 3 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 1 is unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
2	Why cold and dry both work, and that they multiply	1
3	Meiosis and segregation: one copy of each gene into each gamete	2, 4
15	Photosynthesis as where a crop's dry mass comes from	6, 12, 15
16	Where water and mineral nutrients do and do not go	6, 12
17	Resource allocation: defence, growth and yield out of one budget	12, 13, 15, 38
21	Genotype by environment: a result belongs to its conditions	11, 38, 44
26	In situ against ex situ conservation, and what each preserves	2, 13, 20, 23, 27, 28, 36, 39

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects.

TALLY STRONG

stop 4 · CROSS · day 2

Score plants in batches and watch the ratio settle. The format's own subject — when is there enough data to report — is exactly the syllabus item, "reading a count as evidence for a mechanism". A first tray that wobbles past 3 : 1 and then holds is the lesson; a second that lands near 15 : 1 refuses to fit one gene.

SERVES 3 : 1 and 15 : 1 computed · concepts 3, 4, 5

NEEDS authored bins, an unprinted pass mark, and the honest-number rule

SPOT STRONG

stop 39 · POP · day 13

The stop already exists: "leave part of it unsprayed and mark the plants that stay clean." A rogueing crew whose criterion changes from **pull anything flowering early** to **pull anything with rust** is what happens when the season turns.

SERVES concept 21 (G×E) off the select-only list · concept 13

RISK score only the discriminating items

CHAIN STRONG

stop 12 · DRY · day 4

The opening a leaf takes carbon in through is the opening it loses water through — a transfer path with one governing link, which is what CHAIN is. The best CHAIN available in this game, and it lands on a select-only concept.

SERVES concepts 15, 16, 17

NEEDS a per-link reading on every link — the field the importer refuses under any other name

TRIAL STRONG

stop 27 · VAULT · day 9

The rings exist because of isolation distance and contamination direction, and stop 27's takeaway is already "the loss happened in the field, on the way back in." Drive the inspection route: clean nursery before infected plots, vault before field, never the reverse.

SERVES concepts 1, 12, 26 — and the place stops being scenery

RISK gates stood off each door by $d/2 + 10$, or they render under the floor

SWEEP STRONG

stop 22 or 24 · TRIAL

Drag selection intensity against realised gain and the curve bends over: select harder on a noisy trial and you buy scatter, not response. $R = h^2 \cdot S$ is already computed here — this makes the h^2 visible instead of given.

SERVES concept 20, regression to the mean, onto OPERATE

BELT MEDIUM

the threshing floor

A literal threshing floor and a processing hall. The honest bit of subject matter is bagged versus open-pollinated heads — a real sort, done at speed, carrying concept 24.

SERVES concept 24 · and the most grade-6-friendly item here

RISK decidable at a glance, or it is dexterity

HOLD MEDIUM

stop 1 or a DRY stop

Concept 2 — cold and dry both work, and they multiply — is a product, and HOLD is the only format that makes a player feel one: hold seed moisture in a band while wet lots keep arriving.

RISK HOLD's own subject is control engineering. It teaches the biology only if the band closes from the moisture-temperature product. On stop 1 it is also the first thing a player ever does.

Germination shelf by shelf, finding where the pattern breaks. Fun, and concept 1 is already the best-served in the game at CONSTRUCT across seven stops. Buys engagement and no curriculum.

The rings are painted isolation distances, so this is the best physical fit in the catalogue — and LOB withholds the launch speed **so it cannot be computed**, which would teach that isolation distance is a feel. Isolation distances are set from measured data. The fun is real and the lesson is backwards.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs seedbank --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING				
4	d2	CROSS	CHOICE → TALLY	Counting a generation the generation counted, so both ratios are computed
12	d4	DRY	CHOICE → CHAIN	What the crop cannot do without water one opening, two flows, and the link that governs
39	d13	POP	CHOICE → SPOT	An outbreak in the right place a rogueing criterion that changes when the season turns
CHOICE 29 → 26 (58%) · moves 31·12·2 → 28·13·4 · effective formats 2.3 → 2.9 · 2 new operate stops, 1 of them arcade				

Balanced 7 STOPS · THE RECOMMENDATION				
4	d2	CROSS	CHOICE → TALLY	Counting a generation
12	d4	DRY	CHOICE → CHAIN	What the crop cannot do without water
39	d13	POP	CHOICE → SPOT	An outbreak in the right place
6	d2	DRY	CHOICE → BALLPARK	Where a tonne of grain comes from where a tonne of grain comes from — closed as a ledger
15	d5	DRY	CHOICE → ALLOCATE	Short is not simply worse harvest index as one finite budget
22	d8	TRIAL	BALLPARK → SWEEP	What one round returns selection intensity against realised gain
1	d1	VAULT	CHOICE → HOLD	Why the vault is cold cold and dry, as a product you hold — only if the band closes from it
CHOICE 29 → 23 (51%) · moves 31·12·2 → 25·14·6 · effective formats 2.3 → 3.6 · 4 new operate stops, 2 of them arcade				

Ambitious 10 STOPS · ADDS WORLD WORK

4	d2	CROSS	CHOICE → TALLY	Counting a generation
12	d4	DRY	CHOICE → CHAIN	What the crop cannot do without water
39	d13	POP	CHOICE → SPOT	An outbreak in the right place
6	d2	DRY	CHOICE → BALLPARK	Where a tonne of grain comes from
15	d5	DRY	CHOICE → ALLOCATE	Short is not simply worse
22	d8	TRIAL	BALLPARK → SWEEP	What one round returns
1	d1	VAULT	CHOICE → HOLD	Why the vault is cold
27	d9	VAULT	CHOICE → TRIAL	What the store did right the inspection route, driven
33	d11	CROSS	SEQUENCE → BELT	The fortnight when it is possible bagged versus open heads, at threshing-floor speed
43	d15	VAULT	CHOICE → ATTEST	What the catalogue says and what is there what the catalogue says and what is there — the format's own sentence

CHOICE **29 → 21** (47%) · moves **31·12·2 → 23·14·8** · effective formats **2.3 → 4.3** · **6** new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/seedbank.yml seedbank --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check seedbank and npm run drive seedbank, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from themes/seedbank/content/ via engine/dev/curriculumDelivery.mjs; concept and equation lists authored in tools/syllabus.js. Format verdicts and slates are editorial. Run it with THEME=seedbank npm run dev.